Subject: Mathematics | **PERIODIC TEST – 02 (POLYNOMIALS)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 11 and Question 12(iii).
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 2 VSAQ questions of 2 marks each.
4. **Section C** comprises 2 SA questions of 3 marks each.
5. **Section D** comprises 2 LA / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case Study with 3 sub-parts of 5 marks total.

SECTION A: Objective & MCQs (5 × 1 = 5 Marks)

Q1. What should be subtracted from the polynomial $x^2 - 16x + 30$ so that 15 is a zero of the resulting polynomial? [1]

(A) 30 (B) 14 (C) 15 (D) 16

Q2. The zeroes of the quadratic polynomial $x^2 + ax + a$, $a > 0$: [1]

(A) cannot both be positive (B) cannot both be negative
(C) are always unequal (D) are always equal

Q3. The value of x for which the polynomial $p(x) = x^2 - 2x - 8$ has a minimum value is: [1]

(A) 1 (B) -1 (C) 2 (D) -4

Q4. If one zero of the quadratic polynomial $4x^2 - 8kx - 9$ is negative of the other, then $k =$: [1]

(A) 1 (B) -1 (C) 0 (D) 2

Q5. **Assertion (A):** If the sum and product of zeroes of a quadratic polynomial are -3 and 2 respectively, then the polynomial is $x^2 + 3x + 2$.
Reason (R): A quadratic polynomial with sum S and product P is given by $k(x^2 - Sx + P)$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. If α and β are the zeroes of the polynomial $x^2 - 4x + 3$, find the value of $\alpha^4\beta^3 + \alpha^3\beta^4$. [2]

Q7. Find a quadratic polynomial whose zeroes are reciprocals of the zeroes of $f(x) = ax^2 + bx + c$, $a \neq 0, c \neq 0$. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. If α and β are zeroes of $p(x) = 3x^2 + 2x + 1$, find the polynomial whose zeroes are $\frac{1-\alpha}{1+\alpha}$ and $\frac{1-\beta}{1+\beta}$. [3]

Q9. If α and β are the zeroes of $f(x) = x^2 - 1$, find a quadratic polynomial whose zeroes are $\frac{2\alpha}{\beta}$ and $\frac{2\beta}{\alpha}$. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. If α and β are the zeroes of $f(x) = 2x^2 - 5x + 7$, form a quadratic polynomial whose zeroes are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$. [5]

Q11. If α and β are the zeroes of $f(x) = x^2 - 2x + 3$, find a quadratic polynomial whose zeroes are: [5]

(i) $\frac{\alpha-1}{\alpha+1}$ and $\frac{\beta-1}{\beta+1}$ (ii) α^3 and β^3

SECTION E: Applied Case Study Analysis (1 × 5 = 5 Marks)

Question 12: Case Study Analysis — Mountain Railway Tunnel Arch

A single-track mountain railway tunnel has an entrance in the shape of a parabola. The width of the tunnel at ground level is 8 m and its maximum height at the centre is 6 m. Taking the centre of the track on the ground as the origin $(0, 0)$:

(i) Write the coordinates of the two ground contact points of the arch and the apex point. [1]

(ii) Form the quadratic polynomial $y = f(x)$ defining the tunnel entrance arch profile. [2]

(iii) **(a)** Find the height of the tunnel arch at a distance of 2 m from the centre line. [2]

— OR —

(b) A rectangular freight container train of width 4 m (extending 2 m on each side) and height 4.2 m enters the tunnel. Determine with calculations if the container will pass safely. [2]

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